

Supplementary Materials

Individual differences in the computational processes that support face recognition

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Keywords: face perception, memory, familiarity, person knowledge, decision models, evidence accumulation modelling

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Parameter recovery study

Priors

These are the priors used for the preregistered LBA model.

parameter	description	condition	correct	response	mean	sd
v_SNovel:IMFALSE	Drift rate	Novel	False	-	0	2
v_SLearned:IMFALSE	Drift rate	Learned	False	-	0	2
v_SNovel:IMTRUE	Drift rate	Novel	True	-	1	2
v_SLearned:IMTRUE	Drift rate	Learned	True	-	1	2
sv_IMTRUE	SD drift rate (true accumulator)	-	True	-	log(0.5)	0.5
B_IRNovel	Threshold	-	-	Novel	log(1)	0.5
B_IRLearned	Threshold	-	-	Learned	log(1)	0.5
A	Start point	-	-	-	log(0.5)	0.3
t0	Non-decision time	-	-	-	log(0.3)	0.3

Model checks

Convergence

The chains showed satisfactory mixing as evidenced by the rhat values.

Posterior predictive checks

Posterior predictive checks

Figure 1.

Prior vs posterior plots

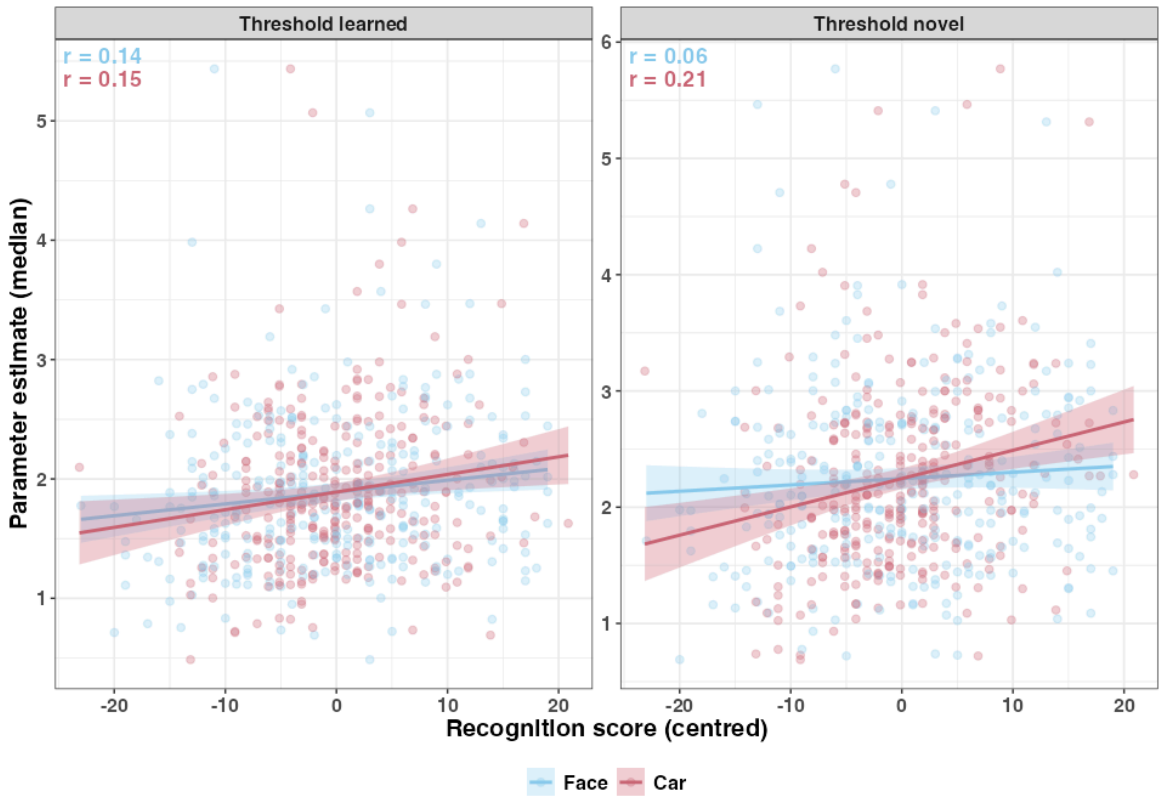
Subject-level drift rates and thresholds

To visualize the individual variability of our parameters of interest, subject-level posterior distributions for threshold (Learned and Novel) and drift rate (Learned and novel) can be found in Figure 2.

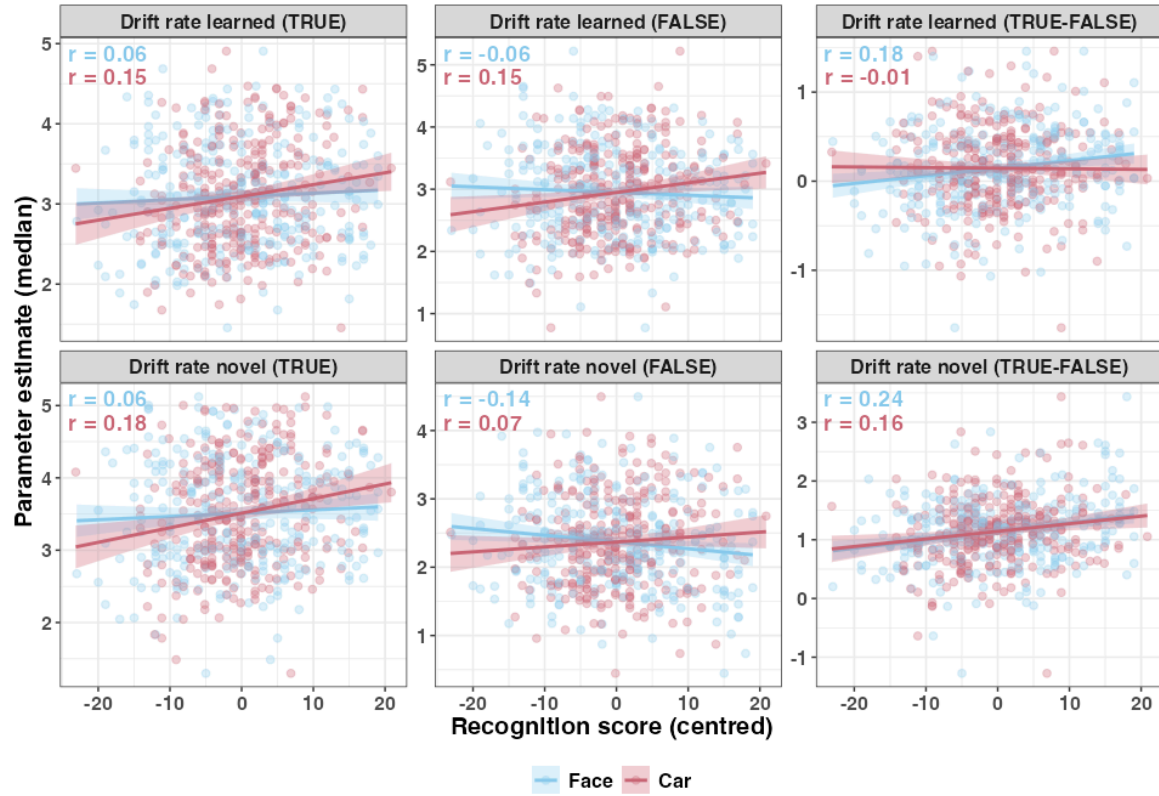


Figure 2. Subject-level posterior distribution of the threshold parameter (Learned and Novel) and the drift rate parameter (Learned and Novel). The top distribution visualize the group-level distribution. Black points denotate the median of the distribuiona and error bars the 95% QI intervals.

**Point estimate correlations with Cambridge Car Memory score and Cambridge
Face Memory scores**



(a) Median threshold correlations with recognition ability



(b) Median drift rate correlations with recognition ability

Figure 3. Point estimates correlations with recognition scores. In (b), drift rate (TRUE) reflects the accumulation of evidence when the response is correct and drift rate (FALSE) the accumulation of evidence when the response is incorrect. Drift rate (TRUE-FALSE) represent the signal-to-noise ratio provided by the stimulus.

Plausible value correlations with Cambridge Car Memory scores and Cambridge Face Memory scores

We calculated the posterior correlations distributions between car recognition ability and our parameters of interest. Correlations with threshold can be found in Figure 4 and correlations with drift rate in Figure 5. Table 1 summarize these distributions. The posterior correlations with face recognition ability were added to facilitate comparisons between the two memory scores.

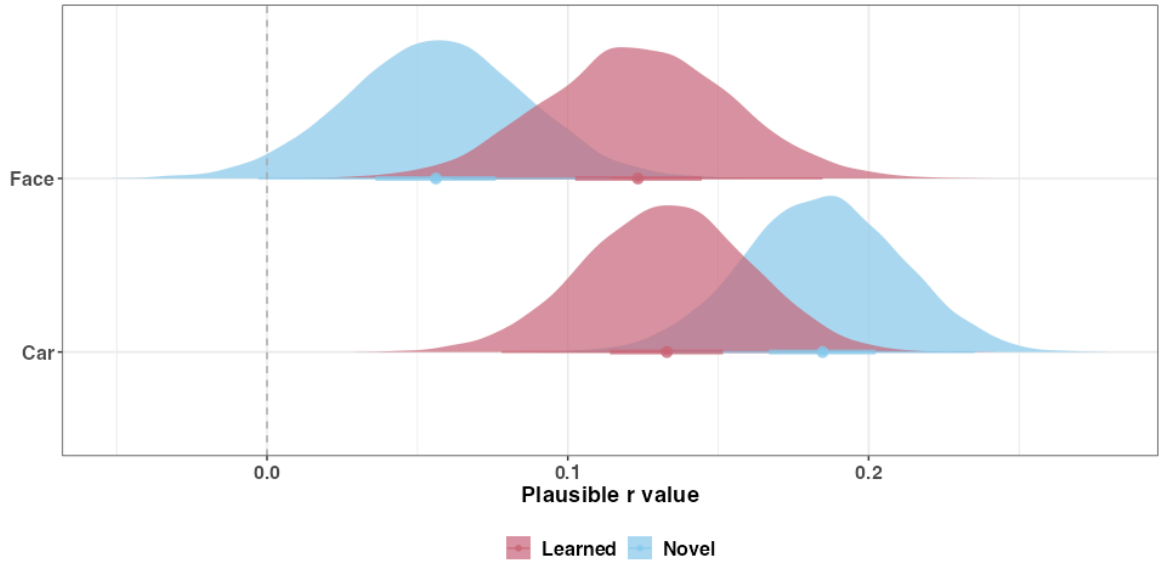
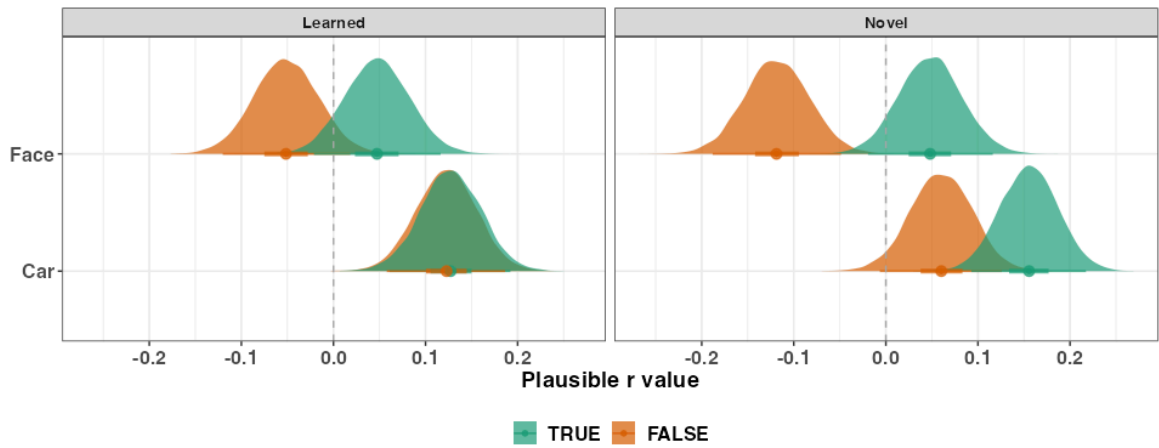
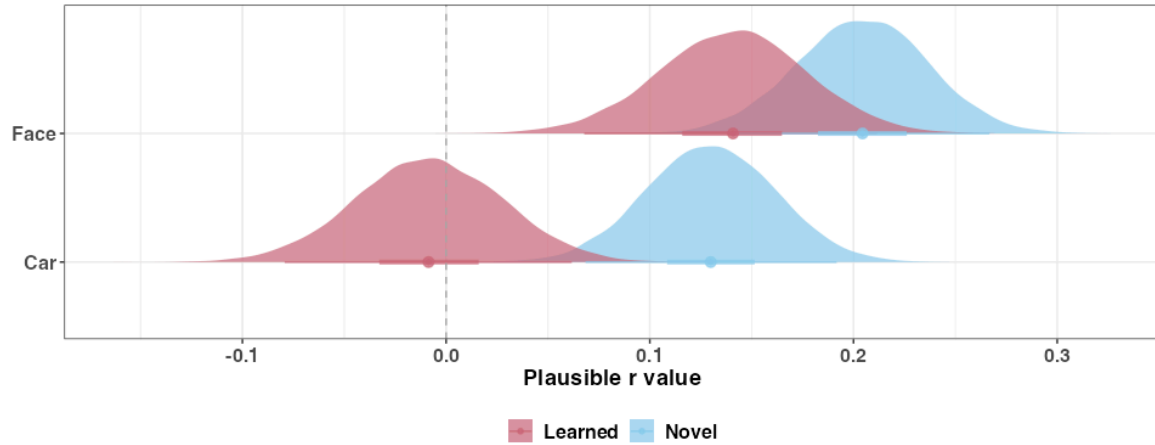


Figure 4. Plausible correlation values distributions between threshold by response and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.



(a) Plausible value correlations between drift rate (TRUE) and drift rate (FALSE) and recognition abilities for both stimulus conditions



(b) Plausible value correlations between drift rate (TRUE-FASLE) and recognition abilities for both stimulus conditions

Figure 5. Plausible correlation values distributions between drift rate parameters by stimulus and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.

Table 1. Median r and 95% quantile intervals of the plausible value correlations between LBA parameters and recognition ability scores.

Parameter	Condition	Accuracy	Face		Car	
			median r_{Face}	95% QI_Face	median r_{Car}	95% QI_Car
Threshold	Learned	-	0.12	[0.06, 0.18]	0.13	[0.08, 0.19]
Threshold	Novel	-	0.06	[-0.00, 0.11]	0.18	[0.13, 0.24]
Drift	Learned	TRUE	0.05	[-0.02, 0.12]	0.13	[0.06, 0.19]
Drift	Learned	FALSE	-0.05	[-0.12, 0.02]	0.12	[0.06, 0.19]
Drift	Learned	TRUE-FALSE	0.14	[0.07, 0.21]	-0.01	[-0.08, 0.06]
Drift	Novel	TRUE	0.05	[-0.02, 0.12]	0.16	[0.09, 0.22]
Drift	Novel	FALSE	-0.12	[-0.19, -0.05]	0.06	[-0.01, 0.13]
Drift	Novel	TRUE-FALSE	0.20	[0.14, 0.27]	0.13	[0.07, 0.19]

Note. Blue = 95% QI entirely above zero; red = 95% QI entirely below zero.

Bayesian linear regression models for threshold parameter

We ran two Bayesian linear regression models to estimate if face recognition ability could predict the threshold parameter (response cautions) for both the Learned and Novel responses. The models description can be found in Table 2.

Table 2. Formula specification for the pre-registered linear regression models.

Parameter	Condition	Ability	Formula
Median threshold	response	Face	Median threshold: $0 + \text{response} + \text{face_ability} + (1 \text{participant})$
Median drift rate	response	Face + Car	Median drift rate: $0 + \text{response} + \text{face_ability} + \text{car_ability} + (1 \text{participant})$

response = two-choices response (Learned vs. Novel)

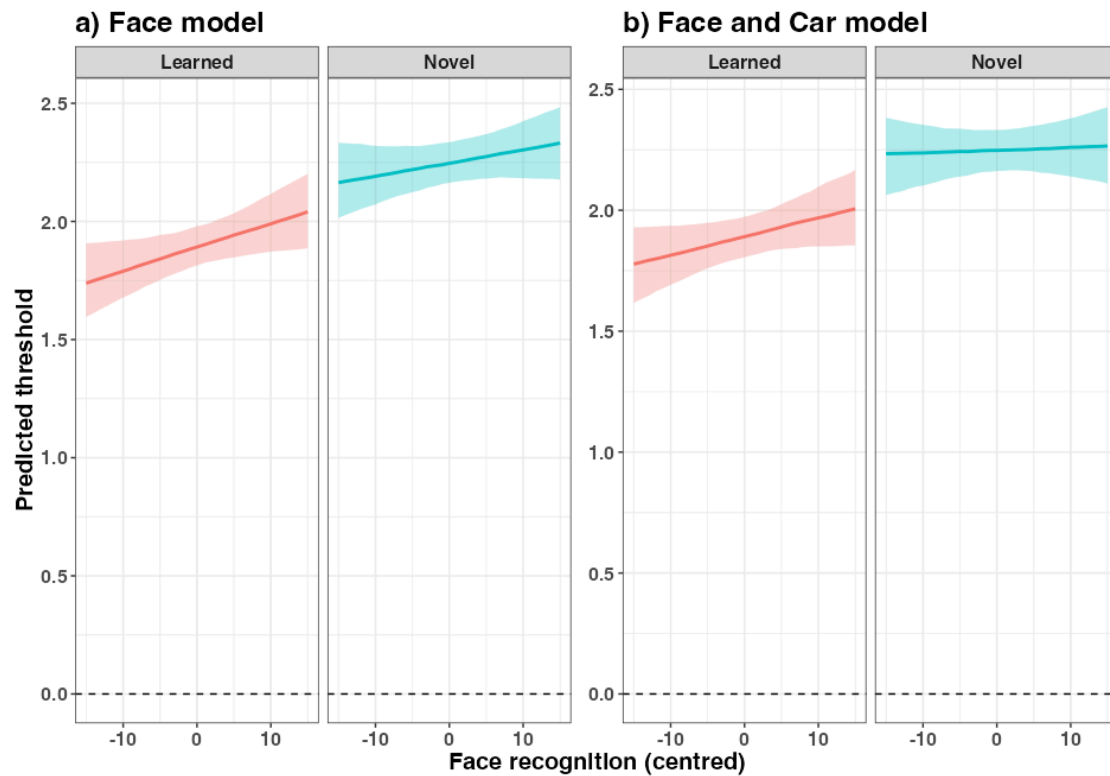


Figure 6. Predicted threshold values from face recognition ability scores for a) Face model and b) Face and Car model

Table 3. Estimates and 95% quantile intervals of the predicted threshold values slope from face recognition ability scores

Parameter	Model	Response	Estimate	l-95% CI	u-95% CI
Threshold slope	Face	Learned	0.0101	0.0012	0.0185
Threshold slope	Face	Novel	0.0056	-0.0032	0.0137
Threshold slope	Face + Car	Learned	0.0076	-0.0005	0.0177
Threshold slope	Face + Car	Novel	0.0012	-0.0078	0.0101

Bayesian linear regression models: Car predictor

As an exploratory analysis, we were interested to see how does object recognition ability predict the drift rate and threshold values from the face recognition task. We ran two Bayesian linear regression models: the first one to estimate if object recognition ability could predict the response caution for learned and novel response and the second one to estimate if object recognition ability could predict the different drift rates (TRUE, FALSE and TRUE-FALSE) for learned and novel face stimuli. The models specificity can be found in Table 4.

Table 4. Formula specification for the pre-registered linear regression models.

Parameter	Condition	Ability	Formula
Median threshold	response	Car	Median threshold: $0 + \text{response} + \text{car_ability} + (1 \text{participant})$
Median drift rate	familiarity:accuracy	Car	Median drift rate: $0 + \text{familiarity:accuracy} + \text{car_ability} + (1 \text{participant})$

Parameter	Condition	Ability	Formula
response = two-choices response (Learned vs. Novel); familiarity = face stimulus familiarity (Learned vs. Novel); accuracy = response accuracy (TRUE vs. FALSE)			

Regarding the threshold (response caution) predictions, visual inspections of the predictions show that as car recognition ability increases, the response caution for both, the Learned and Novel responses, increases (Figure 7).

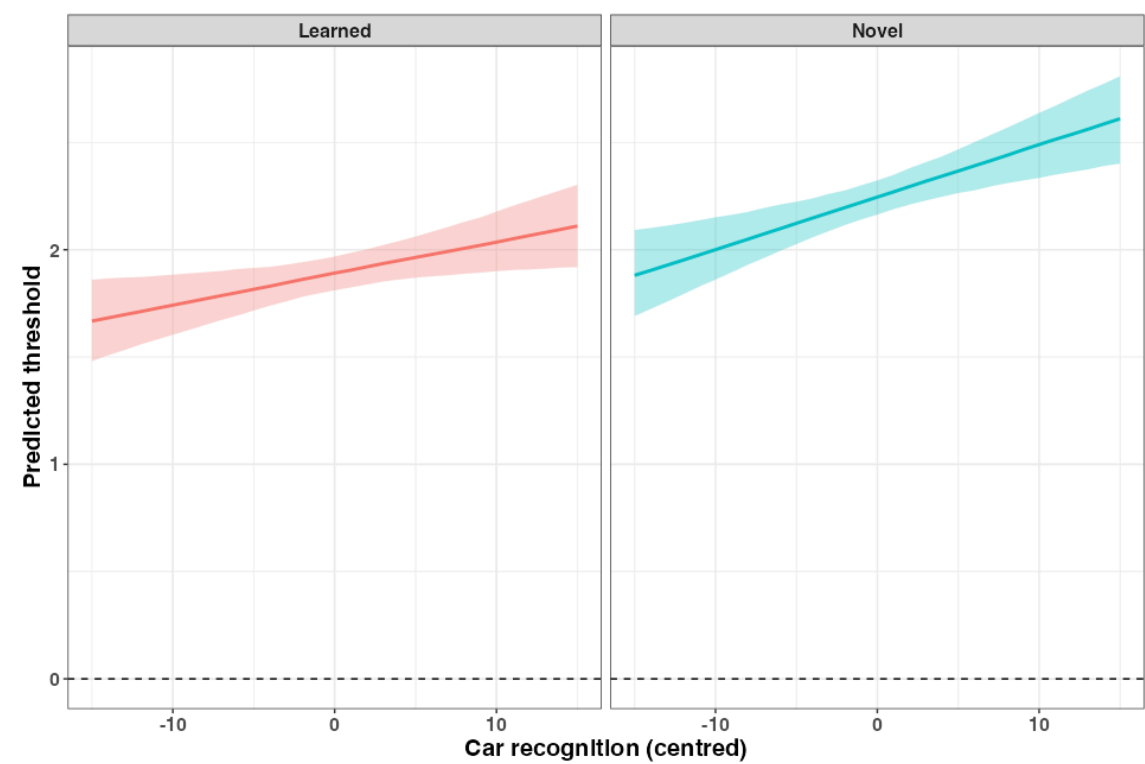


Figure 7. Predicted threshold values from facarce recognition ability scores for learned and novel responses

Regarding the drift rate (TRUE) and (FALSE), visual inspections of the predictions show that, as car recognition ability increases, both drift rate (TRUE and FALSE) for both stimuli conditions (Learned and Novel) increases as well. In summary, as car recognition ability increases, the accumulation of signal (drift rate TRUE) and the accumulation of noise (FALSE) increase, regardless of the stimulus condition (Figure 8).

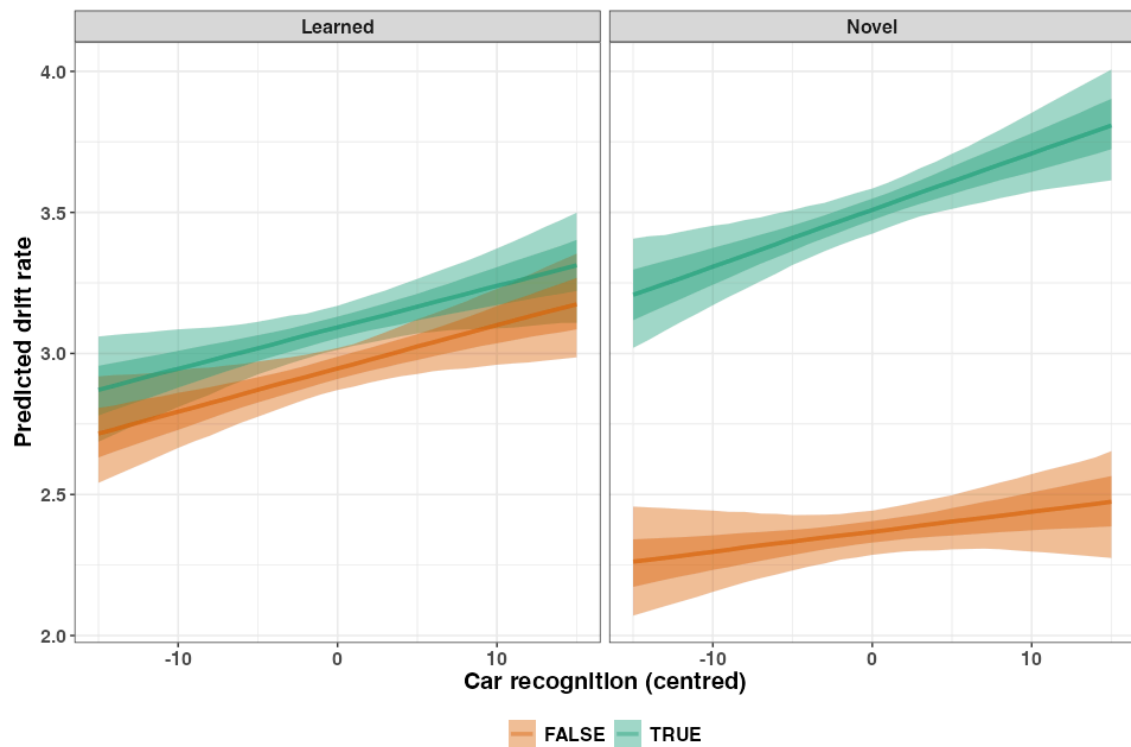


Figure 8. Predicted drift rate (TRUE and FALSE) values from car recognition ability scores for learned and novel faces

Regarding the drift rate (TRUE-FALSE), or the signal-to-noise ratio provided by the stimulus, visual inspections of the predictions show that, as car recognition ability increases, only the drift rate for Novel faces increases, whereas the drift rate for Learned face does not follow this pattern. In summary, as car recognition ability increases, the ability to discriminate Novel faces increases but not the ability to discriminate Learned faces (Figure 9).

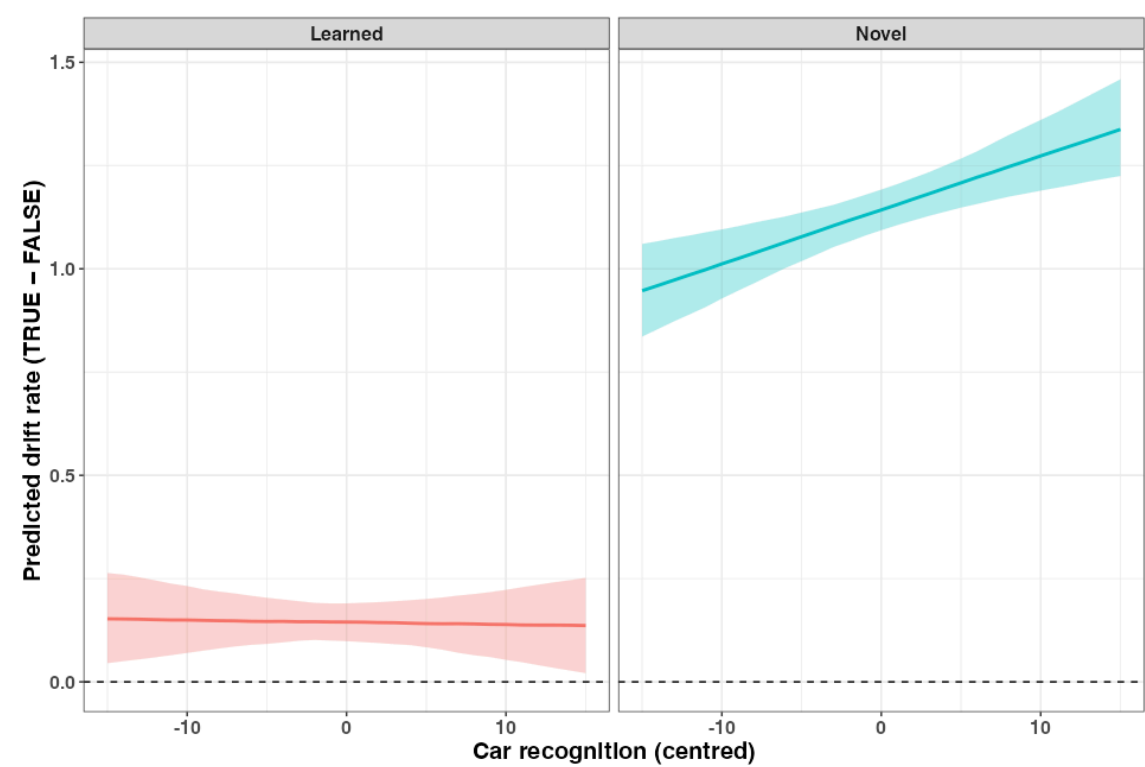


Figure 9. Predicted drift rate (TRUE-FALSE) values from car recognition ability for learned and novel faces